Secure & Redundant Enterprise Network Infrastructure

(Video)

1. Executive Summary

This project demonstrates the end-to-end design, configuration, and deployment of a multi-departmental enterprise network. The architecture is built on a hierarchical model (Access, Core, Edge) prioritizing High Availability (HA), Zero-Trust security principles, and scalable IP management. The environment successfully simulates a corporate headquarters with a secure, redundant connection to the public internet.

2. Network Topology & Architecture

- **Access Layer:** Utilizes Cisco 2960 switches to provide physical connectivity to end-user devices across designated departments.
- **Core Layer:** Powered by a Cisco 3650 Multi-Layer Switch, handling inter-VLAN routing and acting as the centralized DHCP authority.
- **Edge Layer:** Dual Cisco ISR4331 routers configured for failover, providing secure NAT translation and internet routing.

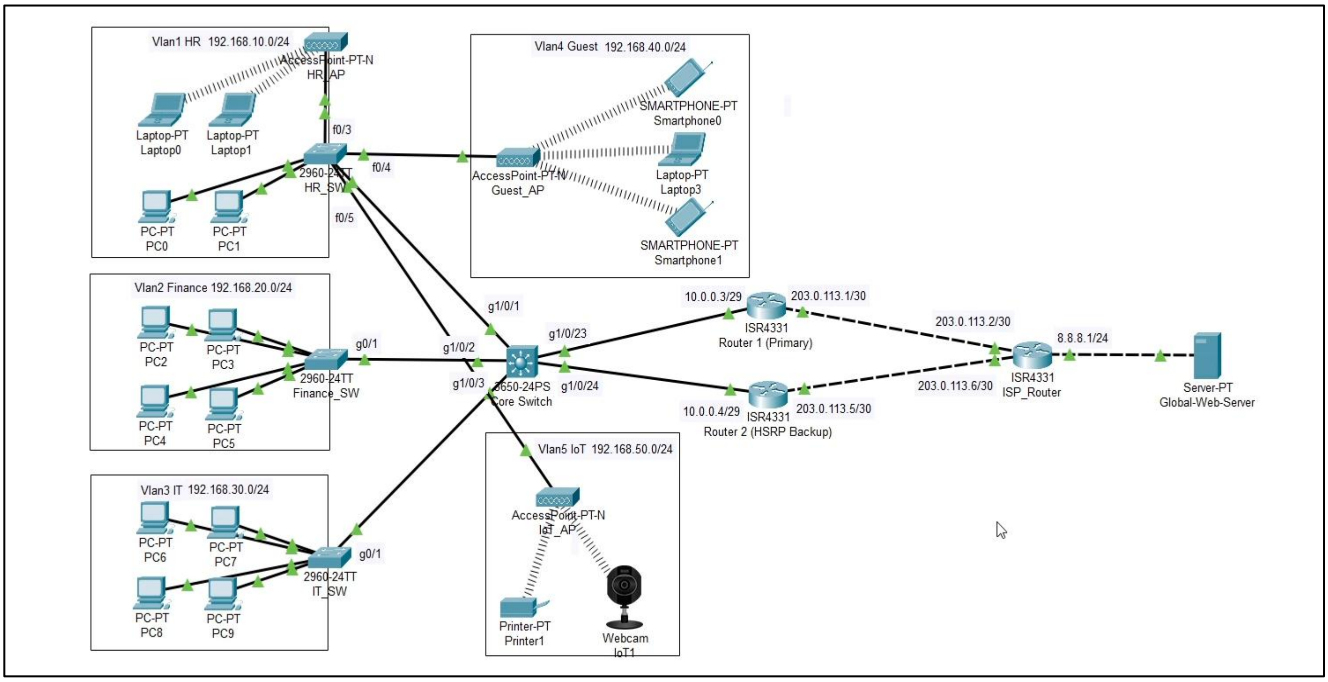

Figure 1: Complete logical and physical topology of the enterprise infrastructure, featuring departmental VLAN segmentation, a multi-layer core, and a redundant HSRP edge.

3. Technical Specifications

A. VLAN & IP Subnet Scheme

The network is logically segmented to reduce broadcast domains and isolate departmental traffic.

VLAN ID	Department	Network Address	Default Gateway (SVI)
VLAN 1	HR	192.168.10.0/24	192.168.10.1
VLAN 2	Finance	192.168.20.0/24	192.168.20.1
VLAN 3	IT	192.168.30.0/24	192.168.30.1
VLAN 4	Guest / Public	192.168.40.0/24	192.168.40.1
VLAN 5	IoT / Devices	192.168.50.0/24	192.168.50.1

B. Core Services

- **DHCP:** Configured on the Core Switch to automatically lease IP addresses, Subnet Masks, and DNS servers to all internal endpoints.
- **DNS:** Simulated external DNS authority mapping the FQDN [*www.global.com*](*https://www.global.com*) to the public IP *8.8.8.8*.
- **NAT / PAT:** Configured "NAT Overload" on the active Edge Router to translate all internal private IP addresses to a single public ISP-facing address.

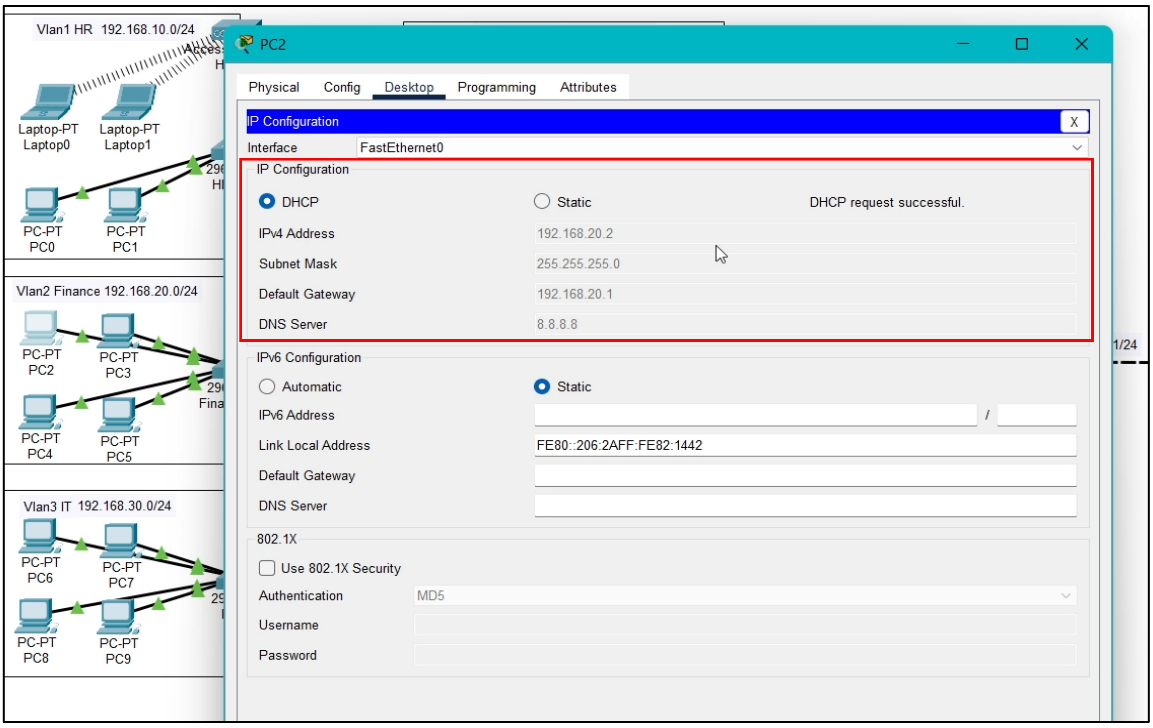

Figure 2: Successful automated IP lease and DNS server assignment verified on a Finance VLAN endpoint via the centralized DHCP server.

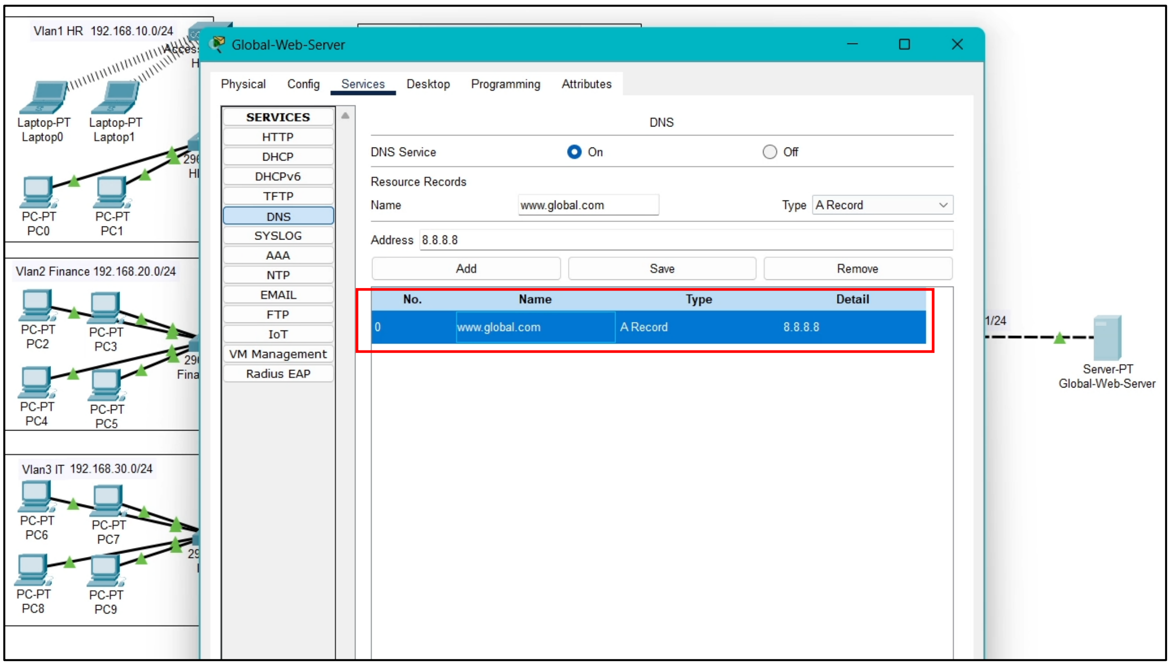

Figure 3: DNS configuration on the simulated public server, mapping the fully qualified domain name to the public IP address (8.8.8.8).

4. Security Implementation

A. Logical Security (Firewall & ACLs)

Implemented a Zero-Trust model for the Guest wireless network using an Extended Access Control List (ACL) on the edge gateway.

- **Rule 1 (Deny):** All traffic originating from the Guest VLAN destined for internal corporate subnets is explicitly dropped.
- **Rule 2 (Permit):** Guest traffic destined for the public internet (*0.0.0.0/0*) is permitted.

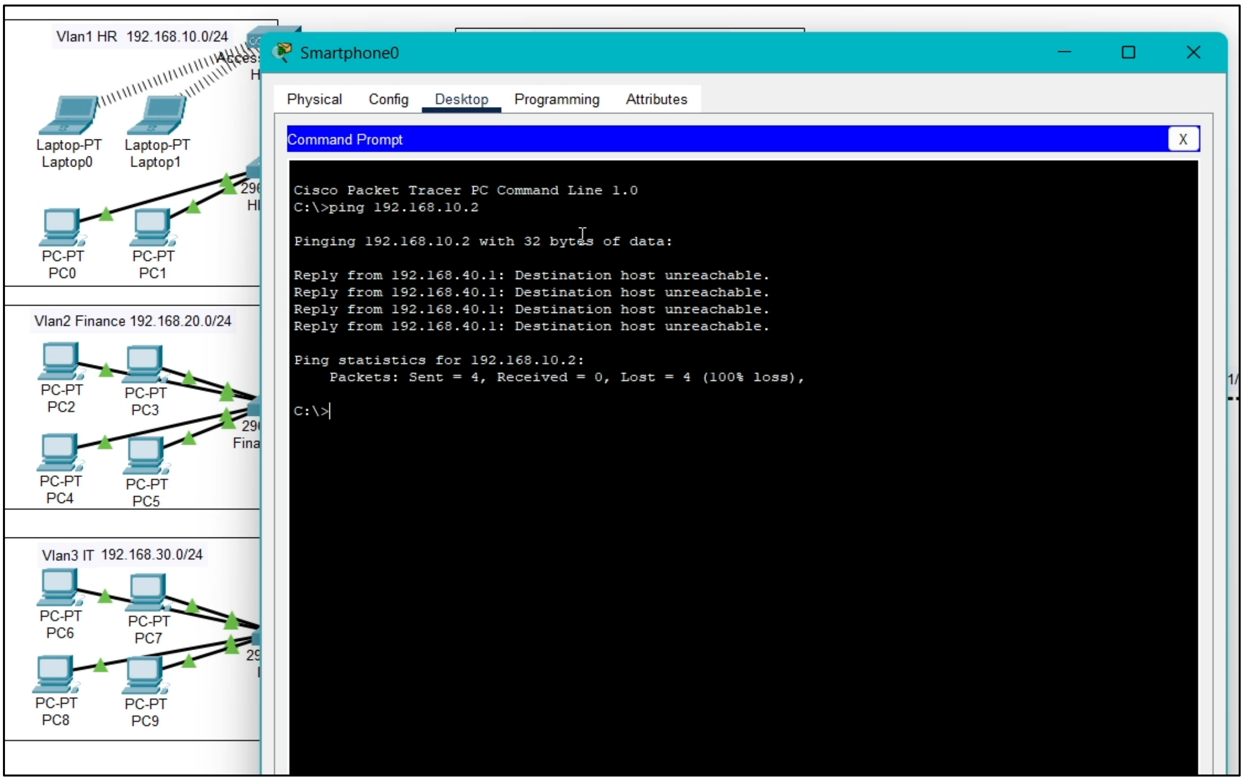

Figure 4: Zero-Trust security verified: Extended ACL successfully blocking a Guest network device from accessing the internal corporate subnet (Destination host unreachable)

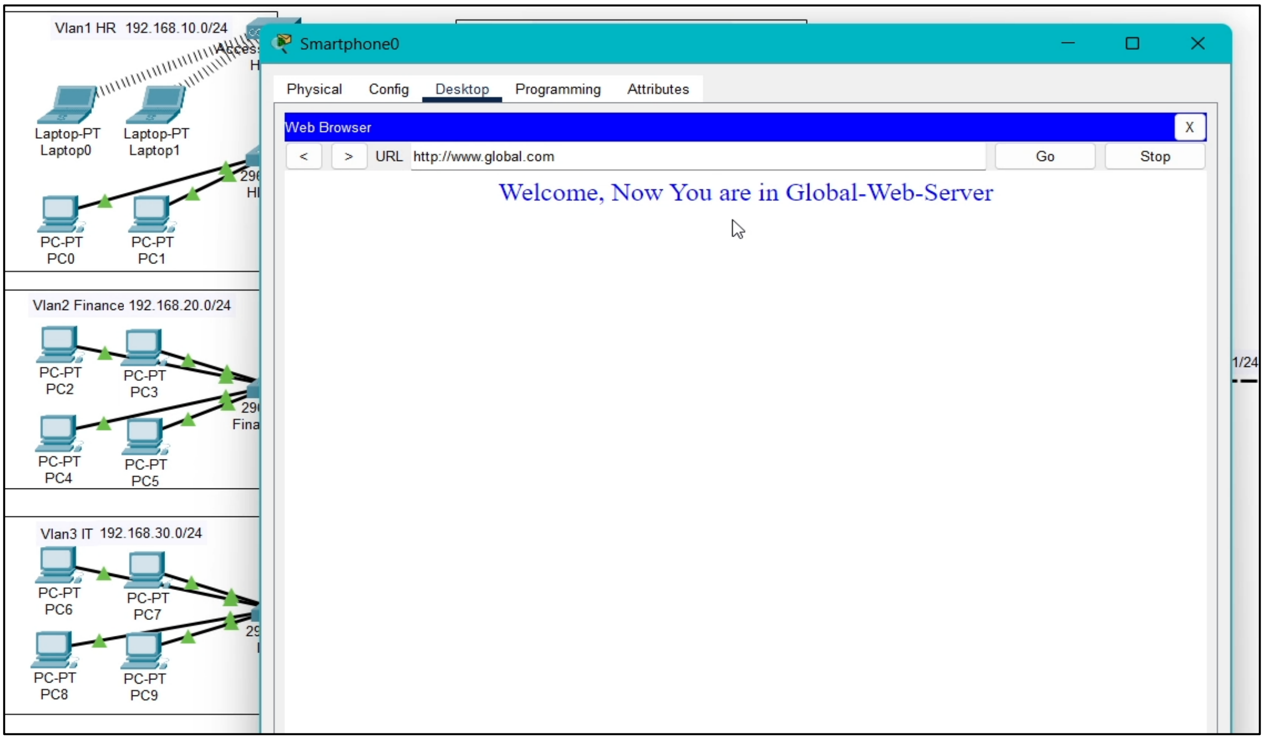

Figure 5: Successful public web server access from the Guest network using a domain name, verifying end-to-end DNS resolution, NAT overload, and outbound routing.

B. Physical Security (Port-Security)

To prevent unauthorized network access and rogue devices, the Access Layer switches were hardened:

- **Wired Endpoints:** Configured with *mac-address sticky* and a maximum limit of 1 device per port.
- **Violation Policy:** Set to *shutdown*. If a rogue device is connected, the port instantly enters an *err-disabled* state.

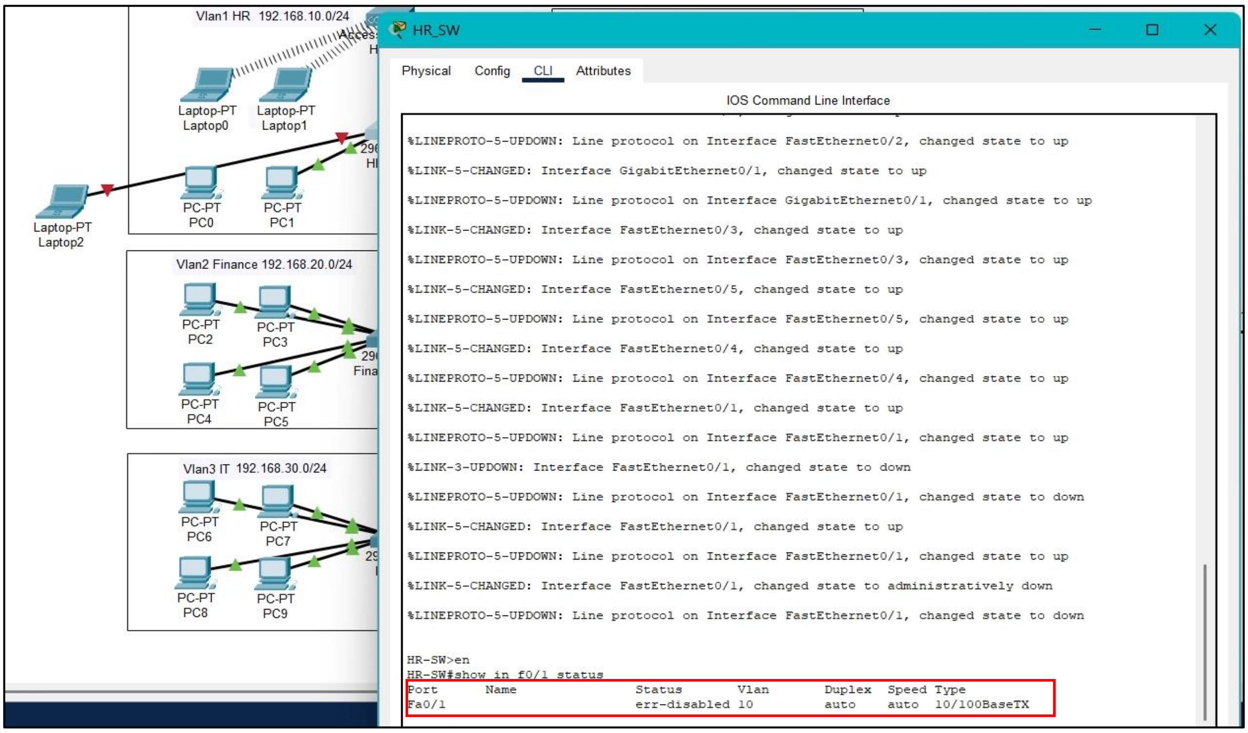

Figure 6: Physical layer security verified: The HR switch interface automatically forced into an *err-disabled* state after detecting an unauthorized rogue device.

5. High Availability & Redundancy

To ensure business continuity, the edge network utilizes HSRP (Hot Standby Router Protocol).

- **Router 1 (Primary):** Configured with a higher priority to actively route all outbound corporate traffic.
- **Router 2 (Standby):** Passively monitors the primary link.
- **Failover Logic:** If the primary ISP link fails, Router 2 automatically assumes the Active Gateway role within 3 seconds, resulting in zero noticeable downtime for internal users.

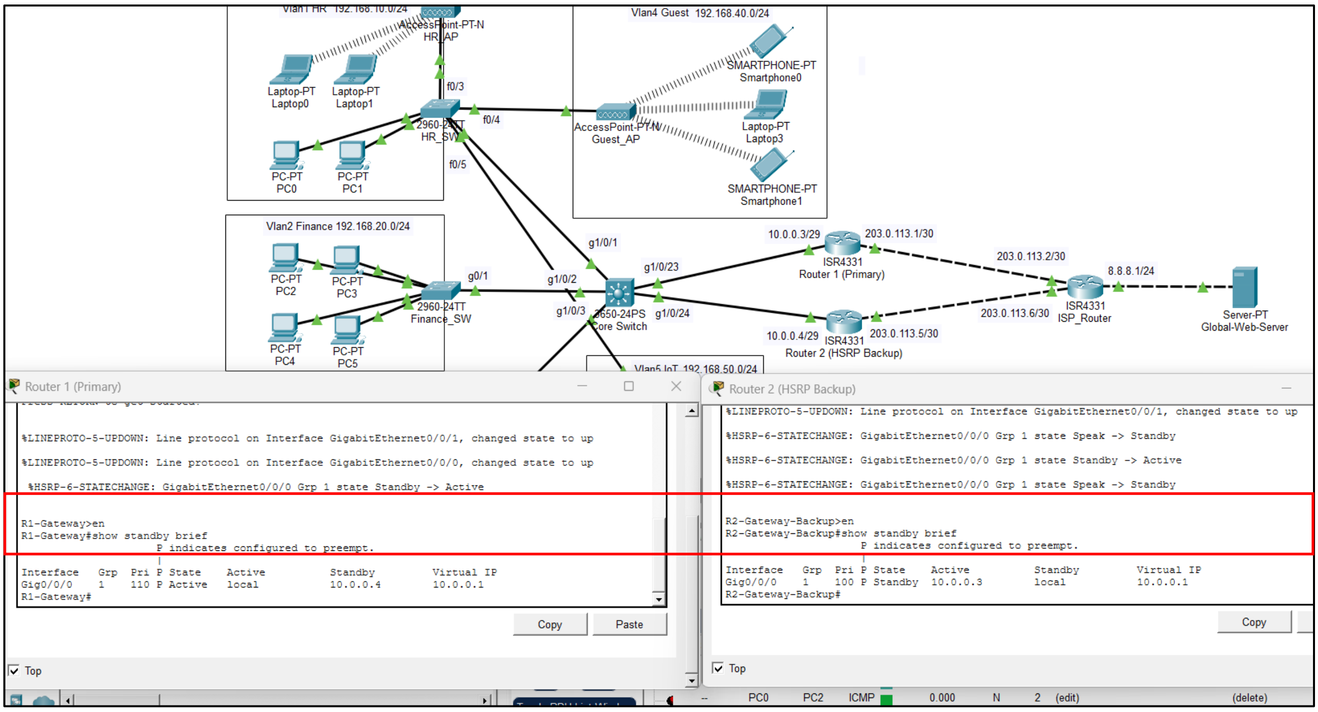

Figure 7: High Availability (HSRP) verification confirming Router 1 as the Active gateway and Router 2 successfully holding the Standby role.

6. Testing & Verification Results

Test Performed	Expected Result	Actual Result	Status
DHCP Distribution	PCs receive correct IP and DNS from Core SW.	Verified via *ipconfig*.	PASS
Inter-VLAN Routing	HR PC can successfully ping IT PC.	ICMP Echo Reply received.	PASS
Internet Resolution	PCs can browse to [*www.global.com*]	Webpage loaded successfully.	PASS
HSRP Failover	Continuous pings survive primary router failure.	Pings resumed after 1-2 drops.	PASS
Port Security Trigger	Rogue PC causes port to enter *err-disabled*.	Link status turned Red (Down).	PASS
